YEAR 9 MATHS · CHAPTER 8 · MIND MAP

Algebraic techniques

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1 Expanding binomial products

Multiply every term in the first bracket by every term in the second.

$(x+3)(x+5) = x^2+8x+15$

$(2x-1)(x+4) = 2x^2+7x-4$

2 Perfect squares and DOTS

$(a+b)^2 = a^2+2ab+b^2$ and $(a-b)(a+b) = a^2-b^2$

$(x+4)^2 = x^2+8x+16$

$(x-5)(x+5) = x^2-25$

Watch out: $(x+3)^2 \neq x^2+9$: don't forget the middle term.

3 Factorising expressions

Take out the highest common factor (HCF).

$6x+9 = 3(2x+3)$ $x^2-7x = x(x-7)$

$-4a-8 = -4(a+2)$

Check by expanding: you should get back to the start.

4 Factorising DOTS

Difference of two squares: $a^2-b^2 = (a-b)(a+b)$

$x^2-49 = (x-7)(x+7)$

$4x^2-9 = (2x-3)(2x+3)$

$3x^2-12 = 3(x-2)(x+2)$

5 Factorising by grouping

Group into pairs, take out a factor from each, then the common bracket.

$x^2+3x+2x+6 = x(x+3)+2(x+3)$

$= (x+3)(x+2)$

6 Monic quadratic trinomials

x^2+bx+c: find two numbers that multiply to c and add to b.

$x^2+7x+12 = (x+3)(x+4)$

$x^2-x-6 = (x-3)(x+2)$

7 Trinomials ax^2+bx+c

Find two numbers that multiply to $a \times c$ and add to b. Split the middle term, then group.

$2x^2+7x+3$: $2 \times 3 = 6$, use 6 and 1

$2x^2+6x+x+3 = 2x(x+3)+1(x+3)$

$= (2x+1)(x+3)$

8 Algebraic fractions: × and ÷

Factorise, cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{x+2}{6} \times \frac{9}{x+2} = \frac{9}{6} = \frac{3}{2}$

$\frac{4x}{5} \div \frac{2x}{15} = \frac{4x}{5} \times \frac{15}{2x} = 6$

9 Algebraic fractions: + and −

Use a common denominator, then add or subtract the numerators.

$\frac{x}{3}+\frac{x}{4} = \frac{4x}{12}+\frac{3x}{12} = \frac{7x}{12}$

$\frac{2}{x}-\frac{1}{2x} = \frac{4}{2x}-\frac{1}{2x} = \frac{3}{2x}$

10 Further simplification

Factorise first, then cancel whole factors.

$\frac{x^2-9}{x+3} = x-3$

$\frac{x^2+5x+6}{x+2} = x+3$

Watch out: cancel factors, never single terms: $\frac{x+3}{3} \neq x$

11 Equations with algebraic fractions

Multiply both sides by the lowest common denominator to clear the fractions.

$\frac{x+1}{3} = \frac{x-2}{2} \Rightarrow 2(x+1) = 3(x-2) \Rightarrow x = 8$

$\frac{x}{2}+\frac{x}{3} = 10$: multiply by 6, $3x+2x = 60$, so $x = 12$